

Q1.

In this question use $g = 10 \text{ m s}^{-2}$.

A particle, of mass 2 kg, is attached to one end of a light elastic string of natural length 0.2 metres.

The other end of the string is attached to a fixed point O .

The particle is pulled down and released from rest at a point 0.6 metres directly below O .

The particle then moves vertically and next comes to rest when it is 0.1 metres below O .

Assume that no air resistance acts on the particle.

- (a) Find the maximum speed of the particle.

(6)

- (b) Describe one way in which the model you have used could be refined.

(1)

(Total 7 marks)

Q2.

An elastic string has one end attached to a point O fixed on a rough horizontal surface. The other end of the string is attached to a particle of mass 2 kg. The elastic string has natural length 0.8 metres and modulus of elasticity 32 newtons.

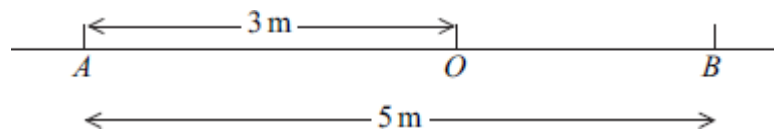
The particle is pulled so that it is at the point A , on the surface, 3 metres from the point O .

- (a) Calculate the elastic potential energy when the particle is at the point A .

(3)

- (b) The particle is released from rest at the point A and moves in a straight line towards O .

The particle is next at rest at the point B . The distance AB is 5 metres.



Find the frictional force acting on the particle as it moves along the surface.

(6)

- (c) Show that the particle does not remain at rest at the point B .

(2)

- (d) The particle next comes to rest at a point C with the string slack.

Find the distance BC .

(2)

- (e) Hence, or otherwise, find the total distance travelled by the particle after it is

released from the point A .

(1)

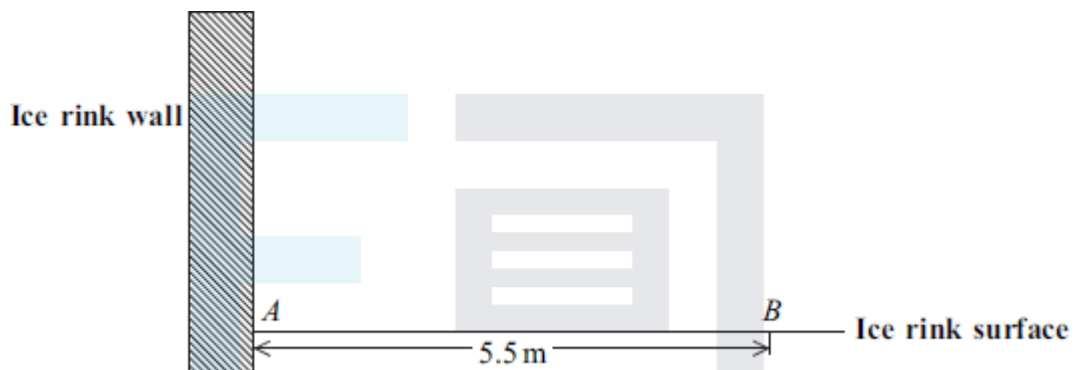
(Total 14 marks)

Q3.

Zoë carries out an experiment with a block, which she places on the horizontal surface of an ice rink. She attaches one end of a light elastic string to a fixed point, A , on a vertical wall at the edge of the ice rink at the height of the surface of the ice rink.

The block, of mass 0.4 kg , is attached to the other end of the string. The string has natural length 5 m and modulus of elasticity 120 N .

The block is modelled as a particle which is placed on the surface of the ice rink at a point B , where AB is perpendicular to the wall and of length 5.5 m .



The block is set into motion at the point B with speed 9 m s^{-1} directly towards the point A . The string remains horizontal throughout the motion.

- (a) Initially, Zoë assumes that the surface of the ice rink is smooth.

Using this assumption, find the speed of the block when it reaches the point A .

(4)

- (b) Zoë now assumes that friction acts on the block. The coefficient of friction between the block and the surface of the ice rink is μ .

- (i) Find, in terms of g and μ , the speed of the block when it reaches the point A .

(6)

- (ii) The block rebounds from the wall in the direction of the point B . The speed of the block immediately after the rebound is half of the speed with which it hit the wall.

Find μ if the block comes to rest just as it reaches the point B .

(6)

(Total 16 marks)

Q4.

A train, of mass 60 tonnes , travels on a straight horizontal track. It has a maximum speed of 40 m s^{-1} when its engine is working at 800 kW .

- (a) Find the magnitude of the resistance force acting on the train when the train is travelling at its maximum speed.

(3)

- (b) When the train is travelling at 40 m s^{-1} , the power is turned off. Assume that the resistance force is constant and is equal to that found in part (a). Also assume that this resistance force is the only horizontal force acting on the train.

Use an energy method to find how far the train travels when slowing from 40 m s^{-1} to 36 m s^{-1} .

(4)

(Total 7 marks)

Q5.

A block, of mass 5 kg , is attached to one end of a length of elastic string. The other end of the string is fixed to a vertical wall. The block is placed on a horizontal surface.

The elastic string has natural length 1.2 m and modulus of elasticity 180 N . The block is pulled so that it is 2 m from the wall and is then released from rest. Whilst taut, the string remains horizontal. It may be assumed that, after the string becomes slack, it does not interfere with the movement of the block.



- (a) Calculate the elastic potential energy when the block is 2 m from the wall.

(2)

- (b) If the horizontal surface is smooth, find the speed of the block when it hits the wall.

(3)

- (c) The surface is in fact rough and the coefficient of friction between the block and the surface is μ .

Find μ if the block comes to rest just as it reaches the wall.

(7)

(Total 12 marks)

Q6.

A light elastic string has natural length 6 metres and modulus of elasticity 300 newtons . It has one end attached to a fixed point, A , on a rough horizontal plane. The other end of the string is attached to a particle of mass 4 kilograms . The particle is pulled along the plane until it is 8 metres from the point A . The particle is then released from rest.

- (a) Calculate the elastic potential energy of the string when the particle is 8 metres from the point A .

(2)

- (b) The coefficient of friction between the particle and the plane is 0.3.

Show that the speed of the particle when the string becomes slack is 6.18 m s^{-1} , correct to three significant figures.

(6)

(Total 8 marks)

Q7.

A lift containing miners moves vertically from rest at one level to rest at a higher level.

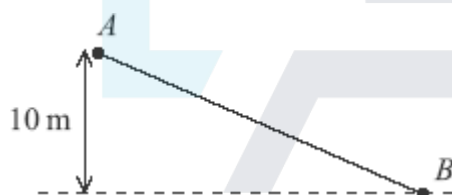
The total mass of the lift and the miners is 800 kg. The vertical distance between the two levels is 200 metres.

Find the work done in raising the lift and the miners from the lower level to the higher level.

(Total 3 marks)

Q8.

A child, of mass 35 kg, slides down a slide in a water park. The child, starting from rest, slides from the point A to the point B , which is 10 metres vertically below the level of A , as shown in the diagram.



- (a) In a simple model, all resistance forces are ignored.

Use an energy method to find the speed of the child at B .

(3)

- (b) State one resistance force that has been ignored in answering part (a).

(1)

- (c) In fact, when the child slides down the slide, she reaches B with a speed of 12 m s^{-1} .

Given that the slide is 20 metres long and the sum of the resistance forces has a constant magnitude of F newtons, use an energy method to find the value of F .

(4)

(Total 8 marks)

Q9.

An elastic string has one end attached to a point O , fixed on a horizontal table. The other end of the string is attached to a particle of mass 5 kilograms. The elastic string has natural length 2 metres and modulus of elasticity 200 newtons. The particle is pulled so that it is 2.5 metres from the point O and it is then released from rest on the table.

- (a) Calculate the elastic potential energy when the particle is 2.5 m from the point O . (2)
- (b) If the table is smooth, show that the speed of the particle when the string becomes slack is $\sqrt{5} \text{ m s}^{-1}$. (3)
- (c) The table is, in fact, rough and the coefficient of friction between the particle and the table is 0.4.
- Find the speed of the particle when the string becomes slack. (7)
- (Total 12 marks)**

Q10.

A block of mass 2 kg is placed on a horizontal surface. An elastic string has natural length 0.5 metres and modulus of elasticity 30 newtons. One end of the string is fixed to the surface at the point O and the other end is attached to the block. The block is pulled along the surface away from O until it is at the point P , where the length of OP is 1.8 metres. The block can be modelled as a particle.

- (a) Calculate the elastic potential energy in the string when the block is at P . (2)
- (b) Assume that the horizontal surface is smooth.
- (i) The block is then released from P and moves towards O . Show that, when the block has moved 0.5 metres, its speed is 5.61 m s^{-1} , correct to three significant figures. (5)
- (ii) Find the speed of the block when it reaches O . (3)
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- (c) Assume that the horizontal surface is rough and that the coefficient of friction between the surface and the block is 0.1. Find the speed of the block when it reaches O . (5)

(Total 15 marks)

Q11.

A ball of mass 0.6 kg is thrown vertically upwards from ground level with an initial speed of 14 m s^{-1} .

- (a) Calculate the initial kinetic energy of the ball. (2)
- (b) Assuming that no resistance forces act on the ball, use an energy method to find the maximum height reached by the ball. (3)
- (c) An experiment is conducted to confirm the maximum height for the ball calculated

in part (b). In this experiment the ball rises to a height of only 8 metres.

- (i) Find the work done against the air resistance force that acts on the ball as it moves. (3)
 - (ii) Assuming that the air resistance force is constant, find its magnitude. (2)
 - (d) Explain why it is **not** realistic to model the air resistance as a constant force. (1)
- (Total 11 marks)**

Q12.

The diagram below shows the cross-section of a slide in a children's playground. Jan, of mass 28 kg, sits at rest on the slide at *A*.

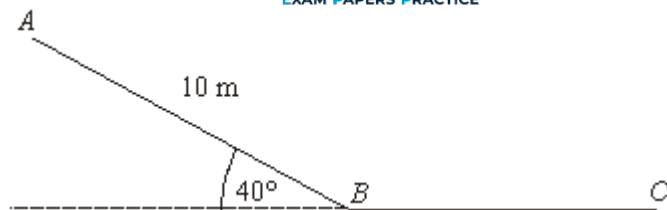


Jan's friend gives her a push which causes her to start moving with speed 1 m s^{-1} . She then slides down the slide to *B*, which is 2.5 metres below the level of *A*.

- (a) In a simple model, Jan's motion is not affected by any forces of resistance.
 - (i) By using energy, show that Jan's speed when she reaches *B* is 7.07 m s^{-1} , correct to three significant figures. (4)
 - (ii) It takes T seconds for Jan to move from *A* to *B*. Sketch a velocity-time graph for Jan's motion as she travels from *A* to *B*. (3)
 - (b) In a more refined model, Jan's motion is assumed to be affected by resistance forces. Given that the work done by the resistance forces while Jan travels from *A* to *B* is 350 J, calculate Jan's speed when she reaches *B*. (4)
- (Total 11 marks)**

Q13.

The diagram shows a simple slide made of two sections *AB* and *BC*. Section *AB* has length 10 metres and is at an angle of 40° to the horizontal. Section *BC* is horizontal.



A child uses the slide, pushing himself off so that at A his speed is 2 m s^{-1} . When he reaches B his speed is 6 m s^{-1} . The mass of the child is 35 kg .

- (a) (i) Calculate the kinetic energy of the child at A . (2)
- (ii) Calculate the gain in kinetic energy as the child moves from A to B . (3)
- (b) Show that the loss in potential energy as the child moves from A to B is 2200 joules, correct to three significant figures. (2)
- (c) As the child moves from A to B , a resistance force acts on him. Assume that this force has a constant magnitude F newtons.
Find F . (3)
- (d) Assume that the same resistance force acts on the child as he moves on BC and that he stops at C . Find the length of BC . (3)
- (Total 13 marks)**

Q14.

A ball has mass 0.5 kg and is released from rest at a height of 6 metres above ground level.

- (a) Assume that no resistance force acts on the ball as it falls.
- (i) Find the kinetic energy of the ball when it has fallen 3 metres. (2)
- (ii) Use an energy method to find the speed of the ball when it hits the ground. (3)
- (b) Assume that a constant resistance force acts on the ball as it falls and that the ball hits the ground travelling at 2 m s^{-1} . Use an energy method to find the magnitude of the resistance force. (4)
- (Total 9 marks)**